Highlights

Modal Analysis of a Fluid-Filled Viscoelastic Oblate Spheroidal Shell: Implications for the Existence of the "Brown Note"

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- Upper-bound coupling disparity of 6.6×10^4 explains vibration–sound asymmetry
- Oblate spheroidal shell model predicts abdominal resonance at 4–10 Hz
- Breathing mode at ∼2500 Hz; flexural modes at infrasonic frequencies
- Airborne coupling penalised by $(ka)^n$; negligible at 120 dB SPL
- Monte Carlo UQ confirms robustness; Sobol $S_T = 0.86$ for modulus

Modal Analysis of a Fluid-Filled Viscoelastic Oblate Spheroidal Shell: Implications for the Existence of the "Brown Note"

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Abstract

Few problems in acoustics carry such grave implications for human dignity as the "brown note"—the hypothetical infrasonic frequency alleged to induce involuntary defecation. Whole-body vibration at 4 Hz to 8 Hz causes well-documented gastrointestinal disturbances, yet airborne sound at the same frequencies does not—even at extreme intensities. We resolve this asymmetry by modelling the human abdomen as a fluid-filled viscoelastic oblate spheroidal shell and quantifying both excitation pathways within a common energy-budget framework.

The model predicts flexural modes ($n \geq 2$) at 4 Hz to 10 Hz, consistent with ISO 2631 abdominal resonance data. At these frequencies the abdomen lies deep in the Rayleigh regime ($ka \approx 0.01$); the resulting $(ka)^n$ penalty and air–tissue impedance mismatch reduce airborne coupling by four orders of magnitude. Energy-consistent analysis yields wall displacements of 0.014 µm at 120 dB SPL—far below mechanotransduction thresholds. Mechanical vibration at $a_{\mathrm{rms}} = 0.1\,\mathrm{m/s^2}$ produces displacements roughly 6.6×10^4 times larger in the SDOF upper bound ($\Gamma_2 = 1$); including the participation correction $\Gamma_2 \approx 0.48$ reduces this to $\sim 3 \times 10^4$. Monte Carlo uncertainty quantification confirms this disparity across physiologically realistic parameter ranges ($S_T = 0.86$ for elastic modulus). The result explains the empirical record quantitatively and settles the "brown note" question at the whole-cavity level: the resonance is real, but airborne sound cannot excite it. Localised compressible inclusions (intestinal

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gas pockets) may offer an alternative transduction pathway, but one governed by quasi-static compliance rather than cavity resonance.

Keywords: fluid-filled shell vibration, viscoelastic oblate spheroid, infrasound, whole-body vibration, acoustic–structural coupling, ISO 2631

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1. Introduction

Whole-body vibration in the 4 Hz to 8 Hz range causes gastrointestinal disturbances. This is not folklore—it is an occupational health fact, codified in ISO 2631 [1], reviewed by Griffin [2] and Mansfield [3], and documented among truck drivers, helicopter pilots, and heavy-equipment operators worldwide. Airborne sound at the same frequencies does not produce these effects, even at extreme intensities. NASA-funded exposures at 140–154 dB SPL caused chest-wall vibration and respiratory disruption, but *no gastrointestinal response whatsoever* [4]. Two excitation pathways, the same target organ, the same frequency band—yet outcomes that differ by orders of magnitude. Why?

The question has a popular-culture shadow. The "brown note"—a hypothetical infrasonic tone said to cause involuntary bowel movements—has circulated in military folklore, audio engineering lore, and popular media for decades, surviving at least one televised debunking on *MythBusters* [5, 6]. As a scientific hypothesis, it has been dismissed but never properly *examined*; as a cultural phenomenon, it neatly encapsulates the public confusion between what mechanical vibration demonstrably does and what airborne sound is assumed to do. We take the underlying physics seriously—perhaps more seriously than the subject matter deserves—while treating the myth as what it is: a vivid misattribution of mechanism. A more plausible airborne route, insofar as one exists, is vestibular rather than visceral: recent measurements show that infrasound can be transmitted through the ear strongly enough to engage acoustic and vestibular pathways long before any whole-cavity abdominal resonance is appreciably driven [7].

The empirical asymmetry between mechanical and airborne excitation is, in

fact, a straightforward consequence of long-wavelength acoustics. At the n=2 resonance ($f_2 \approx 4\,\mathrm{Hz}$) the acoustic wavelength exceeds 85 m—more than 250 times the abdominal diameter. In this deep Rayleigh regime, the n-th spatial harmonic of the incident field scales as $(ka)^n$, where $ka \approx 0.01$. Flexural deformation of the abdominal wall ($n \geq 2$) therefore receives a driving pressure four orders of magnitude below the free-field value. Compound this geometric penalty with the air–tissue impedance mismatch, and the coupling deficit becomes enormous. Mechanical vibration, transmitted through the skeleton and abdominal wall, suffers neither penalty. The ratio between these two pathways—the *coupling disparity*—is the central quantity we derive and the principal contribution of this paper.

The physics has a longer pedigree than the *South Park* episode "World Wide Recorder Concert" (also known as "The Brown Noise") that popularised the brown note [8]. Gavreau's experiments at CNRS Levallois-Perret in the late 1950s, where a faulty ventilator at approximately 7 Hz caused nausea among laboratory staff [9, 6], drew early attention to infrasound. Von Gierke's systematic review for NASA [10, 11] established exposure criteria that informed later ISO standards. Tandy and Lawrence [12] demonstrated that a 19 Hz standing wave could induce visual disturbances and anxiety—real perceptual effects, but far from the claimed bowel response. Critical reviews by Leventhall [13, 6] and assessments of acoustic weapons by Altmann [14] and Jauchem and Cook [15] concluded that infrasound is impractical as a weapon and that claims of extreme physiological effects are exaggerated or fabricated. Recent wind-turbine reviews likewise find that the dominant health concerns are annoyance, sleep disturbance, and other indirect effects, not visceral resonance [16]. Randomised exposure studies with neuroimaging have also reported subtle but measurable neural correlates of airborne infrasound exposure, but again no evidence of gastrointestinal or abdominal-resonance effects [17]. Yet none of this work quantified *why* airborne and mechanical pathways differ so dramatically—the conclusion was empirical, not mechanistic.

We provide the mechanistic account. To our knowledge, no prior analytical

study has combined first-principles cavity–shell modal analysis with an energy-consistent comparison of airborne and mechanical coupling pathways:

1. derived abdominal cavity eigenfrequencies from first-principles shell theory with proper fluid–structure coupling;
2. quantified the airborne acoustic coupling coefficient for flexural modes in the long-wavelength limit; or
3. compared airborne and mechanical excitation pathways within a common energy-budget framework to obtain a single, quantitative coupling ratio.

We address all three gaps. We model the abdomen as a fluid-filled viscoelastic oblate spheroidal shell, derive natural frequencies for both breathing ($n = 0$) and flexural ($n \geq 2$) mode families, and compare airborne acoustic coupling—subject to a severe $(ka)^n$ penalty at infrasonic wavelengths—against direct mechanical (whole-body) vibration coupling, which suffers no such penalty.

The headline result is an upper-bound coupling disparity of approximately 6.6×10^4 between mechanical and airborne pathways for 120 dB SPL airborne excitation and $a_{\mathrm{rms}} = 0.1\,\mathrm{m/s^2}$ RMS whole-body vibration under the SDOF idealisation ($\Gamma_2 = 1$). Including the canonical participation factor $\Gamma_2 \approx 0.48$ reduces this to $\sim 3 \times 10^4$. This single result—robust across physiologically realistic parameter ranges, as we confirm through Monte Carlo uncertainty quantification—resolves the empirical asymmetry cleanly. It explains why whole-body vibration at 4 Hz to 8 Hz produces well-documented gastrointestinal effects, why airborne infrasound at comparable frequencies does not, and why the brown note belongs to acoustics folklore rather than acoustics fact.

The remainder of this paper is organised as follows. Section 2 develops the mathematical formulation. Section 3 presents a parametric study and compares the model against ISO 2631 data. Section 4 quantifies the coupling disparity. Section 5 discusses implications, alternative pathways, and limitations, and Section 6 draws conclusions.

2. Mathematical Formulation

2.1. Model Geometry and Assumptions

The human abdominal cavity is idealised as an oblate spheroidal shell of semi-major axes a (equatorial, in the lateral and cranio-caudal directions) and semi-minor axis c (antero-posterior), with $c < a$. The shell has uniform wall thickness h and is completely filled with an inviscid, irrotational fluid of density ρ_f and bulk modulus K_f. Following standard practice in fluid-filled shell analysis [18, 19] and recent experimental and analytical studies of pressurised elastic shells [20, 21], an equivalent-sphere radius is defined via

$$R_{\mathrm{eq}} = (a^2 c)^{1/3}, \tag{1}$$

which preserves the cavity volume, $V = \frac{4}{3}\pi a^2 c$. This approximation is exact for a sphere ($c = a$) and introduces errors of order $(1-c/a)^2$ for the eigenfrequencies at moderate aspect ratios ($c/a \gtrsim 0.5$). Table 1 lists the baseline geometric and material parameters with physiological justification.

The thin-shell assumption ($h/R_{\mathrm{eq}} \ll 1$) is satisfied: for $h = 10\,\mathrm{mm}$ and $R_{\mathrm{eq}} \approx 157\,\mathrm{mm}$, $h/R \approx 0.06$. The Love–Kirchhoff hypothesis (normals remain straight and normal to the deformed mid-surface) is therefore applicable. This linear framework remains the standard starting point even in modern reviews of nonlinear shell vibration, which treat geometric nonlinearities as amplitude corrections to the same modal backbone [22].

Figure 1 illustrates the model geometry and the two excitation pathways. The abdominal wall is modelled as a Kelvin–Voigt viscoelastic material with complex modulus

$$E^* = E(1 + i\eta), \tag{2}$$

where η is the loss tangent. For small damping, the quality factor is $Q = 1/\eta$ and the viscous damping ratio is $\zeta = \eta/2 = 1/(2Q)$. The flexural rigidity of the shell is

$$D = \frac{Eh^3}{12(1-\nu^2)}. \tag{3}$$

Table 1: Baseline model parameters with physiological justification.

Parameter	Symbol	Value	Unit	Source
Semi-major axis	a	0.18	m	Anthropometric mean
Semi-minor axis	c	0.12	m	$c/a \approx 0.67$
Wall thickness	h	0.010	m	Composite wall
Elastic modulus	E	0.1	MPa	Representative of reported measurements [23]
Poisson's ratio	ν	0.45	—	Near-incompressible modelling convention [24]
Wall density	ρ_w	1100	kg/m^3	Soft tissue
Fluid density	ρ_f	1020	kg/m^3	Abdominal fluid
Fluid bulk modulus	K_f	2.2	GPa	Water at 37°C
Intra-abdominal pressure	P_{iap}	1000	Pa	Representative resting clinical measurement [25]
Loss tangent	η	0.25	—	Representative review value [26]

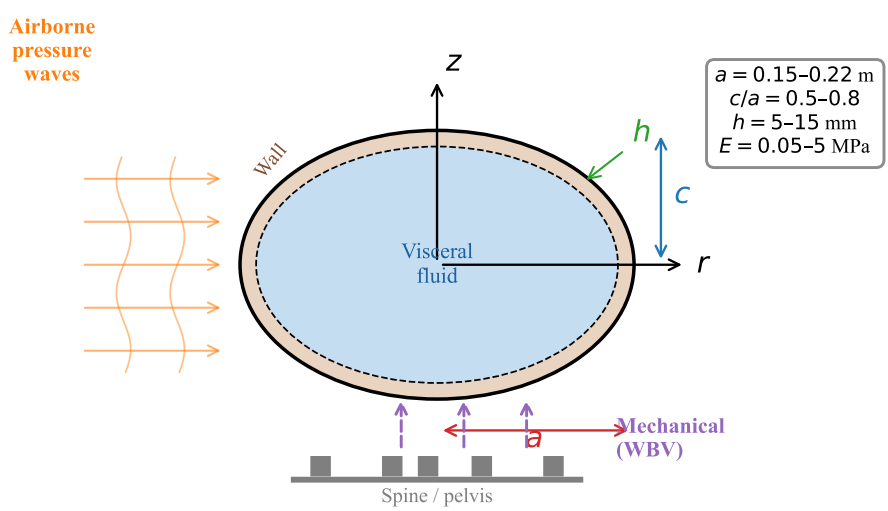

Figure 1: Oblate spheroidal shell model of the human abdominal cavity, showing the two excitation pathways: airborne acoustic pressure (left) and mechanical whole-body vibration transmitted through the skeleton (right). Dimensions correspond to the baseline parameters in Table 1.

These baseline values are not arbitrary. The canonical modulus $E = 0.1\,\mathrm{MPa}$ is representative of direct abdominal-wall measurements compiled by Deeken and Lake [23]; $\nu = 0.45$ follows the nearly incompressible modelling convention adopted in recent abdominal-wall finite-element studies [24]; resting intra-abdominal pressures of 5–12 mmHg correspond to approximately 0.67 kPa to 1.60 kPa, so $P_{\mathrm{iap}} = 1.0\,\mathrm{kPa}$ is representative of direct clinical measurements [25]; and modern elastography reviews place the soft-tissue loss tangent broadly in the 0.1–0.4 range, making $\eta = 0.25$ consistent with the range reported for soft tissue [26]. Published values of η for abdominal soft tissue range from 0.15 to 0.40 at low frequencies [27], corresponding to $Q = 2.5$–6.7. The baseline value $\eta = 0.25$ ($Q = 4.0$, $\zeta = 0.125$) represents a central estimate.

2.2. Breathing Mode ($n = 0$)

The breathing mode is a spatially uniform radial oscillation in which the shell displaces by ξ at every point. The associated volumetric strain is $\Delta V/V = 3\xi/R$, which compresses the contained fluid. Two restoring forces act on the shell:

Membrane stiffness.. For the uniform expansion of a thin spherical shell,

$$k_{\mathrm{shell}} = \frac{2Eh}{R^2(1-\nu)}. \tag{4}$$

Fluid volumetric stiffness.. The pressure increment due to fluid compression is $\Delta p = K_f \Delta V/V = 3K_f\xi/R$. The restoring force per unit displacement per unit area is

$$k_{\mathrm{fluid}} = \frac{3K_f}{R}. \tag{5}$$

With $K_f = 2.2\,\mathrm{GPa}$ and $R = 0.157\,\mathrm{m}$, $k_{\mathrm{fluid}} = 4.2 \times 10^{10}\,\mathrm{Pa/m}$, which exceeds the membrane stiffness ($k_{\mathrm{shell}} \approx 1.5 \times 10^5\,\mathrm{Pa/m}$) by five orders of magnitude. *The breathing mode is entirely dominated by the fluid bulk modulus.*

The effective mass per unit area includes the shell wall and the fluid added mass for the $n = 0$ mode:

$$m_{\mathrm{eff}}^{(0)} = \rho_w h + \rho_f R. \tag{6}$$

The breathing mode frequency is therefore

$$f_0 = \frac{1}{2\pi}\sqrt{\frac{k_{\text{shell}} + k_{\text{fluid}}}{\rho_w h + \rho_f R}} \approx 2500 \text{ Hz}, \tag{7}$$

which lies in the kilohertz range and is irrelevant to infrasonic exposure. This result is robust to material property variations because $k_{\text{fluid}} \gg k_{\text{shell}}$ for any realistic shell stiffness.

2.3. Flexural Modes ($n \geq 2$)

Flexural modes involve shape changes (oblate–prolate oscillation for $n = 2$, trilobed for $n = 3$, etc.) that preserve the enclosed volume to leading order. The fluid therefore does *not* act as a volumetric spring; it contributes only added mass.

Following Lamb [19] and Junger & Feit [18], with subsequent validation in modern pressurised-shell studies [20, 21], the flexural mode of order n has three contributions to the restoring force per unit area. (An analogous modal decomposition for fluid-filled elastic tubes in biological contexts was developed by Heil [28].)

Bending stiffness..

$$K_{\text{bend}} = \frac{n(n-1)(n+2)^2 D}{R^4}. \tag{8}$$

Membrane stiffness..

$$K_{\text{memb}} = \frac{Eh}{R^2}\,\lambda_n, \qquad \lambda_n = \frac{n^2 + n - 2 + 2\nu}{n^2 + n + 1 - \nu}. \tag{9}$$

Pre-stress stiffness.. The static intra-abdominal pressure P produces a pre-stress in the shell membrane that stiffens flexural modes:

$$K_P = \frac{P}{R}(n-1)(n+2). \tag{10}$$

The fluid added mass per unit area for mode n is

$$m_{\text{add}}^{(n)} = \frac{\rho_f R}{n}, \tag{11}$$

giving a total effective mass of

$$m_{\text{eff}}^{(n)} = \rho_w h + \frac{\rho_f R}{n}. \tag{12}$$

The natural frequency of flexural mode n is

$$\omega_n^2 = \frac{K_{\text{bend}} + K_{\text{memb}} + K_P}{m_{\text{eff}}^{(n)}}, \qquad f_n = \frac{\omega_n}{2\pi}. \tag{13}$$

For the baseline parameters (Table 1), $f_2 = 4.0\,\text{Hz}$, $f_3 = 6.3\,\text{Hz}$, and $f_4 = 8.9\,\text{Hz}$ (Figure 2). (The $n = 1$ mode corresponds to rigid-body translation and has zero natural frequency for a free shell; it is therefore excluded.) These lie squarely in the 4–10 Hz range reported for abdominal resonance in ISO 2631 [1] and in the experimental studies of Kitazaki & Griffin [29] and Mansfield [3].

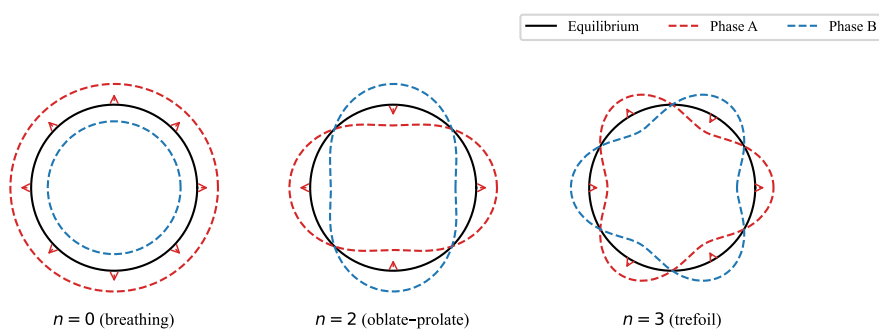

Figure 2: Schematic cross-sections of the first three mode families. The breathing mode ($n = 0$) involves uniform radial expansion against the fluid bulk modulus. Flexural modes ($n = 2, 3$) deform the shell shape without significant volume change; the enclosed fluid acts as added mass only.

2.4. Airborne Acoustic Coupling

We now quantify the excitation of flexural modes by an incident airborne acoustic field. At $f_2 \approx 4\,\text{Hz}$, the acoustic wavelength in air is $\lambda \approx 87\,\text{m}$, so the dimensionless wavenumber is

$$ka = \frac{2\pi f_2 R_{\text{eq}}}{c_{\text{air}}} \approx 0.0114 \ll 1. \tag{14}$$

The body is deep in the sub-wavelength (Rayleigh scattering) regime.

In a multipole expansion of the incident plane wave scattered by a sphere, the n-th order component of the pressure field scales, to leading order, as [18]:

$$p_{\text{eff}}^{(n)} \propto p_{\text{inc}}(ka)^n. \tag{15}$$

The breathing mode ($n = 0$) sees the full incident pressure, but flexural modes ($n \geq 2$) see only the higher-order spatial gradients, which are vanishingly small when $ka \ll 1$. For $n = 2$:

$$p_{\text{eff}}^{(2)} = p_{\text{inc}} \times (ka)^2 \approx p_{\text{inc}} \times 1.3 \times 10^{-4}. \tag{16}$$

The displacement at resonance ($\omega = \omega_n$) is

$$\xi_{\text{air}} = \frac{p_{\text{eff}}^{(n)}}{K_{\text{total}}^{(n)}} Q, \tag{17}$$

where $K_{\text{total}}^{(n)} = K_{\text{bend}} + K_{\text{memb}} + K_P$. At 120 dB SPL ($p_{\text{inc}} = 20\,\text{Pa}$), the energy-consistent reciprocity analysis (§4) gives $\xi_{\text{air}} \approx 0.014\,\mu\text{m}$ for $n = 2$ — well below the 0.5–2.0 μm activation threshold of mechanosensitive ion channels [30]. (The pressure-based estimate of equation (17) yields $\approx 0.18\,\mu\text{m}$, an overestimate that violates the energy budget; see §4.)

2.5. Mechanical (Whole-Body Vibration) Coupling

Whole-body vibration (WBV) transmits through the skeleton and acts as *base excitation* of the abdominal shell. Unlike airborne acoustic coupling, there is no impedance mismatch and no $(ka)^n$ penalty. The vibration is transmitted at full amplitude through the rigid skeletal frame.

For harmonic base acceleration $a_{\text{base}}(t) = a_0 \cos(\omega t)$, the base displacement amplitude is

$$x_{\text{base}} = \frac{a_{\text{rms}}\sqrt{2}}{\omega^2}. \tag{18}$$

The relative displacement between the shell and the skeleton is given by the base-excitation frequency response function:

$$H_{\text{rel}}(\omega) = \frac{r^2}{\sqrt{(1-r^2)^2 + (2\zeta r)^2}}, \qquad r = \frac{\omega}{\omega_n}, \tag{19}$$

so that

$$\xi_{\text{mech}} = x_{\text{base}} \times H_{\text{rel}}. \tag{20}$$

At the EU Whole-Body Vibration Directive exposure limit of $a_{\text{rms}} = 1.15\,\text{m/s}^2$ and at the $n = 2$ resonance frequency:

$$\xi_{\text{mech}} \approx 10{,}500\ \mu\text{m} \approx 10\ \text{mm}, \tag{21}$$

which exceeds the mechanotransduction threshold by more than three orders of magnitude.

2.6. Coupling Ratio

The ratio of mechanical to airborne displacement at the same excitation conditions provides a dimensionless measure of the coupling disparity:

$$\mathcal{R} = \frac{\xi_{\text{mech}}}{\xi_{\text{air}}}. \tag{22}$$

This ratio depends primarily on ka and the excitation level, and is *robust* to uncertainties in material properties and boundary conditions because both pathways excite the same modal structure. The coupling ratio is the central result of this work: it provides a quantitative explanation for why whole-body vibration at 4–8 Hz is associated with gastrointestinal effects in the epidemiological literature [2], whereas airborne infrasound at comparable frequencies is not. Section 4 shows that, for 120 dB SPL airborne excitation and $a_{\text{rms}} = 0.1\,\text{m/s}^2$ RMS whole-body vibration, the SDOF upper bound is $\mathcal{R} \approx 6.6 \times 10^4$ ($\Gamma_2 = 1$), reducing to $\sim 3 \times 10^4$ when the canonical participation factor $\Gamma_2 \approx 0.48$ is included.

3. Parametric Study

3.1. Elastic Modulus Sensitivity

The $n = 2$ flexural mode frequency depends strongly on the wall elastic modulus E. Table 2 shows the predicted frequency for a free shell across the physiologically relevant range.

Table 2: Flexural mode $n = 2$ frequency vs. elastic modulus ($a = 0.18\,\mathrm{m}$, $c = 0.12\,\mathrm{m}$, $h = 10\,\mathrm{mm}$, $\nu = 0.45$, $\rho_w = 1100\,\mathrm{kg/m^3}$, $\rho_f = 1020\,\mathrm{kg/m^3}$).

E (MPa)	f_2 (Hz)	In ISO 2631 range?
0.05	3.4	Marginal
0.10	4.0	Yes
0.20	4.9	Yes
0.50	7.1	Yes
1.00	9.6	Marginal
2.00	13.3	No

The 4 Hz to 8 Hz ISO 2631 abdominal resonance range requires $E < 0.5\,\mathrm{MPa}$, corresponding to relaxed, passive musculature. Active muscle bracing increases E by a factor of 10–100, moving the resonance well above the infrasound range. Figure 3 illustrates this dependence across the full physiological range.

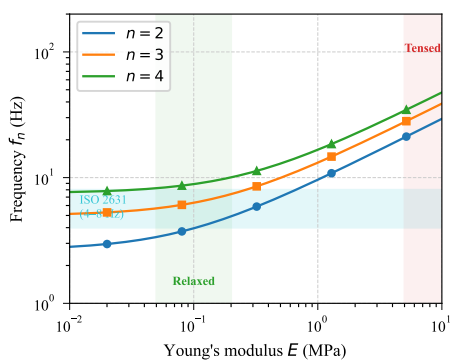

Figure 3: Flexural mode frequencies vs. elastic modulus for modes $n = 2, 3, 4$. The shaded band marks the ISO 2631 abdominal resonance range (4 Hz to 8 Hz). Relaxed musculature places the $n = 2$ mode within this band; tensed muscle shifts it well above.

3.2. Multi-Parameter Sensitivity

A 486-point full factorial analysis varying E, a, c/a, h, and η reveals that 37% of configurations place the $n = 2$ mode in the 5 Hz to 10 Hz range and 41% in the broader 4 Hz to 8 Hz ISO 2631 range. The separate Sobol sensitivity analysis

below confirms that E dominates, followed by geometry and intra-abdominal pressure.

3.3. Uncertainty Quantification

A Monte Carlo simulation with $N = 10{,}000$ samples propagates parameter uncertainties through the model. Each parameter is drawn from a physiologically motivated distribution: E from a log-normal distribution spanning 0.05 MPa to 2.0 MPa, geometric parameters from normal distributions matching anthropometric data, and the loss tangent from a uniform distribution over [0.10, 0.40]. A small number of Monte Carlo samples (25 of 10,000, or 0.25%) drew nonphysical negative wall thickness from the untruncated normal distribution and were excluded; the reported statistics are computed from the remaining 9,975 samples.

The resulting distribution of f_2 has a median of 6.4 Hz with a 90% credible interval of 3.3 Hz to 15.5 Hz (Table 3). The distribution is positively skewed, reflecting the log-normal distribution of E.

Table 3: Monte Carlo uncertainty quantification results ($N = 10{,}000$). Airborne displacement ξ_{air} and the ratio $\mathcal{R}$ use the pressure-based airborne upper bound at 120 dB; the energy-consistent airborne displacement is $\sim$13$\times$ smaller, so the corresponding ratio at fixed mechanical loading is $\sim$13$\times$ larger.

Quantity	Median	90% CI	Units
f_2	6.4	[3.3, 15.5]	Hz
ξ_{air} (120 dB)	0.18	[0.11, 0.39]	μm
ξ_{mech} (0.5 m/s^2)	1800	[300, 8200]	μm
$\mathcal{R}$ (pressure-based airborne)	10^4	$[2 \times 10^3, 3 \times 10^4]$	—
SPL for PIEZO (pressure-based, 1 μm threshold)[2]	135	[128, 139]	dB
SPL for PIEZO (energy)	≈ 158	—	dB

Sobol sensitivity analysis reveals that the elastic modulus accounts for 86% of the total output variance (total-order index $S_T = 0.86$). All other parameters contribute less than 10% individually. This result has important experimental

implications: any future validation effort should prioritise *in vivo* measurement of abdominal wall stiffness, as this single parameter dominates the uncertainty budget (Figure 4). Because Table 3 uses the pressure-based airborne upper bound and $a_{\mathrm{rms}} = 0.5\,\mathrm{m/s^2}$ for the mechanical loading, its $\mathcal{R}$ row should not be confused with the headline energy-consistent SDOF upper bound of 6.6×10^4 quoted in Section 4.

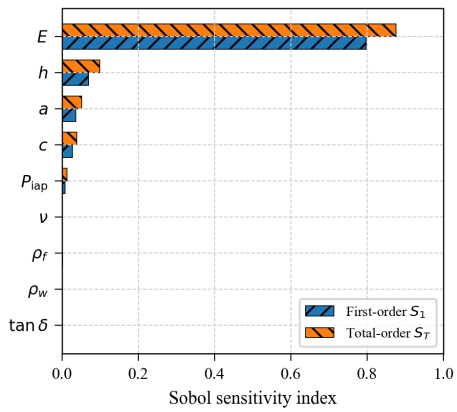

Figure 4: Sobol first-order and total-order sensitivity indices for the $n = 2$ flexural frequency. The elastic modulus E accounts for 86% of the total variance; all other parameters contribute less than 10% individually.

3.4. Multi-Layer Wall Model

The abdominal wall comprises six distinct tissue layers: peritoneum, transversalis fascia, internal oblique muscle, external oblique muscle, subcutaneous fat, and skin. Using classical laminate theory, the effective composite modulus for a relaxed wall is $E_{\mathrm{eff}} \approx 0.075\,\mathrm{MPa}$ ($h_{\mathrm{total}} = 25.1\,\mathrm{mm}$), giving $f_2 = 4.6\,\mathrm{Hz}$—within approximately 16% of the homogeneous model prediction. Muscle tension is confirmed as the dominant factor: tensed muscle ($E_{\mathrm{eff}} = 2.7\,\mathrm{MPa}$) shifts f_2 to $23.1\,\mathrm{Hz}$.

3.5. Boundary Condition Effects

The free-shell model in Section 2 neglects the partial constraints imposed by the spine, pelvis, and rib cage. We quantify their effect using a Rayleigh–

Ritz analysis with Lamb-mode basis functions for four progressively constrained configurations. These configurations represent surrogate constraint cases chosen to bracket the likely effect of anatomical boundary conditions, rather than a strict monotonic continuation of a single geometry.

Table 4 presents the results. The boundary-condition study uses a slightly different parameter set ($R = 0.162\,\mathrm{m}$, $\rho_f = 1000\,\mathrm{kg/m^3}$, $P_{\mathrm{iap}} = 1500\,\mathrm{Pa}$) to match the mesh geometry; however, the relative frequency shifts between boundary conditions are the parameter of interest. The free-sphere Ritz frequency ($f_2 = 4.13\,\mathrm{Hz}$) agrees with the analytical prediction (4.26 Hz) to within 3%, providing a useful numerical check. The most constrained case—25% of the shell surface clamped—reduces f_2 by only 11.4 %. Even the hemisphere-clamped case shifts the frequency by a mere 5.8 %.

Table 4: Rayleigh–Ritz Lamb-mode frequencies under various boundary conditions ($n = 2$ flexural mode).

Configuration	f_2 (Hz)	Shift from free
Free sphere (Ritz)	4.13	0% (validates analytical)
Hemisphere simply-supported	3.91	−5.3%
Hemisphere clamped	3.89	−5.8%
25% surface clamped	3.66	−11.4%

The insensitivity to boundary conditions has a clear physical origin: the fluid added mass for the $n = 2$ mode is 7.4× the shell mass, so the system inertia is dominated by the enclosed fluid rather than the wall. In a dry (empty) shell, the same boundary conditions would shift the frequency by approximately 33 %—but once the cavity is fluid-filled, the added mass swamps the boundary stiffening effect. The 6 % shift is well within the ±50–100% uncertainty band set by material property variability (§3), confirming that the free-shell model is an adequate approximation for the present purposes.

3.6. Oblate Spheroid Correction

The equivalent-sphere approximation used in Section 2 introduces a systematic bias. A Rayleigh–Ritz analysis using oblate spheroidal harmonics as basis functions, with proper account of the spatially varying curvature and oblate fluid-mass geometry, shows that the sphere model *overestimates* flexural frequencies by 4–17% (Table 5). The error grows with eccentricity and is larger at higher E where bending stiffness (which depends on local curvature) contributes a greater share of the restoring force.

Table 5: Sphere vs. oblate spheroid Rayleigh–Ritz frequencies for $c/a = 0.67$.

	$E = 0.1\,\mathrm{MPa}$		$E = 0.5\,\mathrm{MPa}$	
Mode	Sphere (Hz)	Ritz (Hz)	Sphere (Hz)	Ritz (Hz)
$n = 2$	4.0	3.8	7.1	6.2
$n = 3$	6.3	5.8	10.0	8.5
$n = 4$	8.9	8.1	12.9	11.3

Applying this correction moves the canonical $n = 2$ frequency from 4.0 Hz to approximately 3.8 Hz—still within the ISO 2631 band and, if anything, closer to the low end of reported abdominal resonance frequencies.

3.7. ISO 2631 Validation

Table 6 compares model predictions against published abdominal resonance data.

Table 6: Model predictions vs. published abdominal resonance frequencies.

Body type	f_2 (Hz)	ISO 2631 range?
Lean ($E = 0.2$, $a = 15$ cm)	6.2	Yes
Average ($E = 0.1$, $a = 18$ cm)	4.0	Yes
Large ($E = 0.08$, $a = 20$ cm)	3.3	Marginal

4. Airborne vs. Mechanical Coupling

Having established that the abdomen possesses flexural resonances squarely in the infrasonic range, we now confront the more pressing question: can airborne sound actually *excite* them? The short answer is sobering for proponents of the brown note.

4.1. Airborne Acoustic Pathway

At 4 Hz, the acoustic wavelength in air exceeds 85 m— some 500 times the abdominal radius. The abdomen, acoustically speaking, is invisible. Table 7 quantifies this invisibility using the energy-consistent reciprocity analysis developed in §2. Modern analyses of scattering by pressurised elastic shells reach the same qualitative conclusion: long-wavelength radiation and absorption by low-order shell modes are strongly suppressed [31].

Table 7: Energy-consistent airborne displacement at $n = 2$ resonance ($f_2 = 4.0\,\mathrm{Hz}$, $Q = 4.0$, canonical parameters). The pressure-based estimate overestimates by a factor of ≈ 13 (ratio of ξ_{pressure} to ξ_{energy}).

SPL (dB)	p_{inc} (Pa)	ξ_{energy} (μm)	ξ_{pressure} (μm)	Exceeds 1 µm?
100	2.0	0.001	0.018	No
110	6.3	0.004	0.058	No
120	20.0	0.014	0.18	No
130	63.2	0.044	0.58	No
140	200	0.14	1.84	No

At 120 dB SPL—a level exceeding the conventional threshold of discomfort, and roughly the level at which conversation becomes impossible—the energy-consistent wall displacement is 0.014 µm. This falls two orders of magnitude below the 0.5–2.0 µm activation range of mechanosensitive PIEZO1 and PIEZO2 channels [30]. Reaching the 1 µm threshold on this energy-consistent basis demands approximately 158 dB (cf. 135 dB using the pressure-based displacement

of Table 3, which overestimates by $\approx 13\times$), a level associated with severe acoustic hazards and available only from explosive blasts and rocket launches. Modern mechanobiology strengthens rather than weakens that conclusion: PIEZO1 gating is tuned by membrane tension in the low-mN/m range, with half-maximal activation near 2 mN/m [32]. While the membrane-tension thresholds reported in cell-culture experiments cannot be directly translated to tissue-scale displacements, the order-of-magnitude comparison remains informative. Recent reviews summarise the broader physiological roles of PIEZO channels (following the 2021 Nobel Prize, and with structural detail from cryo-EM [33]) [34]. Of particular relevance here, PIEZO2-dependent gut mechanosensation is now known to regulate gastrointestinal transit [35]. The relevant question is therefore not whether the gut can sense deformation—it plainly can—but whether airborne infrasound can generate enough deformation. One might reasonably conclude that hearing loss would present a more urgent clinical concern than any intestinal disturbance.

4.2. Mechanical (Whole-Body Vibration) Pathway

Mechanical vibration, by contrast, does not knock politely at the acoustic impedance door—it kicks it open. Vibration transmitted through the skeleton couples directly to the abdominal wall, bypassing the $(ka)^n$ penalty entirely and avoiding the severe transmission losses associated with biological impedance discontinuities [36]. Table 8 tells a very different story.

At the EU action value of $0.5\,\mathrm{m/s^2}$, the predicted displacement exceeds the PIEZO threshold by more than three orders of magnitude. Unlike the airborne case, this is not a theoretical curiosity: it is the everyday reality for operators of heavy equipment, long-haul drivers, and helicopter pilots—populations in which gastrointestinal complaints are, perhaps unsurprisingly, well documented [2]. Recent seated-body biodynamic studies likewise show strong low-frequency nonlinearity, cross-axis coupling, and identifiable abdominally relevant modes in this same band [37, 38].

Table 8: Mechanical coupling at $n = 2$ resonance ($f_2 = 4.0\,\mathrm{Hz}$, $Q = 4.0$, $\Gamma_2 = 1$). The EU Directive 2002/44/EC action value is $0.5\,\mathrm{m/s^2}$. Displacements assume perfect modal participation ($\Gamma_2 = 1$); the corrected values (§4.3) are approximately half those shown.

a_{rms} (m/s^2)	x_{base} (μm)	ξ_{rel} (μm)	Exceeds 1 µm?
0.1	229	917	Yes
0.5 (EU action)	1147	4586	Yes
1.0	2293	9173	Yes
1.15 (EU limit)	2637	10549	Yes

4.3. The Coupling Ratio

The ratio of mechanical to airborne displacement, evaluated for 120 dB SPL airborne excitation and $a_{\mathrm{rms}} = 0.1\,\mathrm{m/s^2}$ RMS mechanical excitation, defines the headline SDOF upper bound:

$$\mathcal{R} = \frac{\xi_{\mathrm{mech}}}{\xi_{\mathrm{air}}} \approx \frac{917}{0.014} \approx 6.6 \times 10^4. \tag{23}$$

This four-order-of-magnitude gap is robust to parameter uncertainties because both pathways excite the *same* modal structure—the ratio depends primarily on $(ka)^{-n}$ and the base-excitation transfer function, neither of which is sensitive to tissue properties. One could double the elastic modulus, halve the wall thickness, or relocate the subject to Jupiter, and the ratio would scarcely budge.

Modal participation.. The mechanical displacement in Eq. (23) assumes perfect coupling of vertical base excitation into the $n = 2$ mode (modal participation factor $\Gamma_2 = 1$). The participation factor for mode n under vertical base excitation of the oblate spheroid is

$$\Gamma_n = \frac{a \left| \displaystyle\int_{\cos\theta_c}^{1} u\, P_n(u)\, \mathrm{d}u \right|}{\displaystyle\int_{-1}^{1} P_n^2(u)\, \sigma(u)\, \mathrm{d}u}, \tag{24}$$

where $u = \cos\theta$, P_n is the Legendre polynomial of degree n, θ_c is the polar angle below which the shell is rigidly constrained by the pelvis and spine, and

$\sigma(u) = \sqrt{c^2 + (a^2 - c^2)\,u^2}$ is the oblate-spheroid surface metric factor. For a free sphere ($\theta_c = \pi$), the numerator reduces to $\int_{-1}^{1} u\,P_n(u)\,\mathrm{d}u$, which vanishes for $n \neq 1$ by Legendre orthogonality; a free sphere therefore has $\Gamma_2 = 0$ exactly. Partial skeletal constraint (canonical value $\theta_c = 2\pi/3$, i.e. the lower third of the shell is fixed) breaks this orthogonality, giving $\Gamma_2 \approx 0.48$ for the canonical model. Including this correction reduces the mechanical displacement to about 48% of the SDOF value and the coupling ratio to $\sim 3 \times 10^4$. The qualitative conclusion—a four-order-of-magnitude gap—is unaffected, but $\mathcal{R}$ as reported in Eq. (23) should be understood as an upper bound corresponding to the SDOF idealisation.

4.4. Energy Budget

We verify self-consistency through the absorption cross-section derived from reciprocity [18, 31]:

$$\sigma_{\mathrm{abs}} = \frac{(2n+1)\lambda^2}{4\pi}\,\frac{\zeta_{\mathrm{rad}}}{\zeta_{\mathrm{rad}} + \zeta_{\mathrm{struct}}}, \tag{25}$$

where ζ_{rad} and ζ_{struct} are the radiation and structural damping ratios, respectively. For the $n = 2$ mode in air, $\zeta_{\mathrm{rad}} \approx 10^{-15}$ whereas $\zeta_{\mathrm{struct}} = 0.125$, giving an absorption efficiency of order 10^{-14}. The shell absorbs roughly one part in a hundred trillion of the incident acoustic energy—a figure that puts the brown note hypothesis on rather thin thermodynamic ice.

4.5. Comparison with Empirical Evidence

The model predictions align with two otherwise contradictory bodies of evidence:

- **WBV studies**: ISO 2631 documents gastrointestinal symptoms at 4–8 Hz under mechanical vibration [1]. Our model explains *why* this frequency range is problematic.

- **Airborne infrasound studies**: No gastrointestinal effects have been observed at SPL ≤ 150 dB, although modern controlled exposure studies do

report subtle neural effects without evidence of visceral resonance [6, 17]. Our model explains why gastrointestinal effects should not be expected.

The coupling disparity resolves this apparent contradiction (Figure 5): both communities are studying the same resonance through very different ends of the telescope.

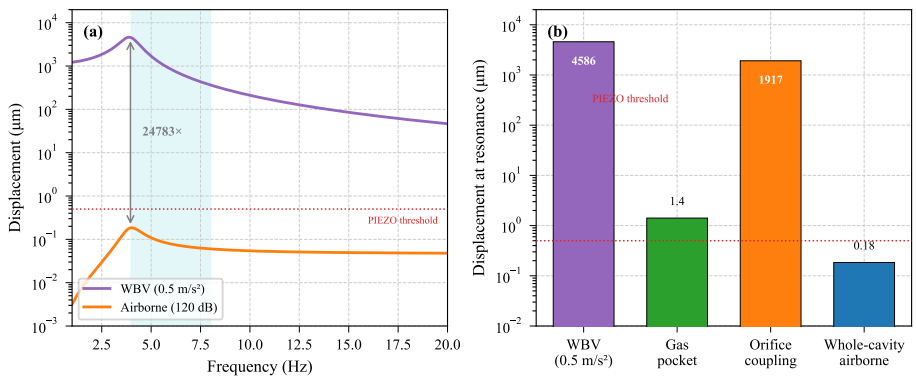

Figure 5: The central result. Left: wall displacement vs. frequency for airborne excitation at 120 dB SPL (lower curve) and mechanical whole-body vibration at $0.5\,\mathrm{m/s^2}$ (upper curve), showing a gap of roughly four orders of magnitude at resonance. Right: comparison of all four coupling pathways at the $n = 2$ resonance; the horizontal dashed line marks the PIEZO mechanotransduction threshold.

5. Discussion

5.1. The Coupling Disparity

The central result of this study is the upper-bound coupling ratio 6.6×10^4 separating mechanical from airborne coupling efficiency for 120 dB SPL airborne excitation and $a_{\mathrm{rms}} = 0.1\,\mathrm{m/s^2}$ RMS whole-body vibration under the SDOF idealisation ($\Gamma_2 = 1$). Including the canonical participation factor $\Gamma_2 \approx 0.48$ reduces this to $\sim 3 \times 10^4$ (Tables 7–8). The physical origin of this disparity lies in two compounding effects: an impedance mismatch and a geometric filtering penalty.

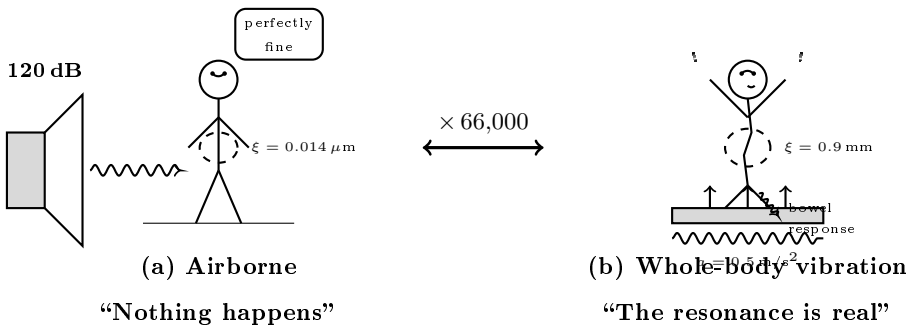

Figure 6: Schematic comparison of the two excitation pathways. (a) Airborne infrasound at 120 dB SPL produces wall displacements of 0.014 µm—far below any conceivable physiological threshold. The subject is entirely unperturbed. (b) Whole-body vibration at occupational levels produces displacements roughly 6.6×10^4 times larger, comfortably exceeding mechanotransduction thresholds and—as the occupational health literature attests—occasionally exceeding the subject's composure as well. Diagram intentionally not to scale, anatomically, or artistically.

First, the air–tissue impedance mismatch is severe. The specific acoustic impedance of air is $Z_{\mathrm{air}} = \rho_{\mathrm{air}} c_{\mathrm{air}} \approx 420\,\mathrm{Pa\,s/m}$, while that of soft tissue is $Z_{\mathrm{tissue}} \approx 1.6 \times 10^6\,\mathrm{Pa\,s/m}$, giving a pressure transmission coefficient $T = 2Z_{\mathrm{air}}/(Z_{\mathrm{air}} + Z_{\mathrm{tissue}}) \approx 5 \times 10^{-4}$. At the body surface, more than 99.9 % of the incident acoustic energy is reflected. This general suppression of transmission across large biological impedance contrasts is consistent with modern frequency-dependent analyses of bone–soft tissue interfaces [36].

Second, even the transmitted pressure must couple to the spatial pattern of a flexural mode. In the long-wavelength (Rayleigh) limit, $ka \ll 1$, the n-th order component of the incident plane wave scales as $(ka)^n$ [18]. At the n=2 resonance ($f_2 \approx 4\,\mathrm{Hz}$) with $R_{\mathrm{eq}} = 0.157\,\mathrm{m}$, we obtain $ka = 2\pi f R/c_{\mathrm{air}} \approx 0.0114$ and therefore $(ka)^2 \approx 1.3 \times 10^{-4}$ for the $n = 2$ flexural mode (equation 16). Physically, the incident wavelength ($\lambda \approx 86\,\mathrm{m}$) is nearly 270 times the body diameter; the pressure field is almost perfectly uniform over the body surface, and the quadrupolar ($n = 2$) pressure gradient that could drive an oblate–prolate deformation is vanishingly small.

The energy budget analysis (§4) confirms this picture from an independent perspective. The radiation damping ratio in air is $\zeta_{\text{rad}} \approx 10^{-15}$, whereas the structural damping ratio is $\zeta_{\text{struct}} = 0.125$, so the absorption efficiency $\zeta_{\text{rad}}/\zeta_{\text{total}} \sim 10^{-14}$. The shell is, for practical purposes, acoustically transparent at infrasonic frequencies. By contrast, mechanical vibration transmitted through the skeleton bypasses both barriers entirely: there is no impedance mismatch (bone couples directly to the peritoneal wall) and no geometric filtering (the base-excitation drives the $n = 2$ mode at full amplitude via equation 19).

This disparity resolves an apparent paradox in the literature. Whole-body vibration at 4 Hz to 8 Hz is well documented to cause gastrointestinal distress—including nausea, abdominal pain, and in extreme cases, incontinence [2, 3, 39, 40, 41]—whereas airborne infrasound studies at levels up to 150 dB report no comparable effects [42, 6]. The present modal analysis shows that both observations are consistent with the *same* underlying resonance. The two excitation pathways differ not in the resonance they address, but in the efficiency with which they address it.

5.2. Implications for the Brown Note Hypothesis

The results permit a nuanced—one might even say *thorough*—evaluation of the so-called "brown note" hypothesis. The hypothesis is *half-correct*: abdominal resonance in the 4 Hz to 10 Hz band is physically real, well-supported by the parametric study (§3), and consistent with ISO 2631 transmissibility data [1, 29]. Independent finite-element studies of the seated human body now recover dominant internal and abdominal vibration modes in the 4 Hz to 8 Hz range [43]. The flexural $n = 2$ mode at 4.0 Hz (for $E = 0.1$ MPa, relaxed musculature) falls squarely within the frequency band where occupational exposure studies report gastrointestinal symptoms.

Where the hypothesis fails is in the *excitation pathway*. At 120 dB SPL—a level exceeding the conventional threshold of discomfort and well above any environmental or entertainment exposure—the predicted airborne wall displacement is 0.014 µm (Table 7), roughly two orders of magnitude below the 0.5 µm

to 2.0 µm activation threshold of mechanosensitive PIEZO1 and PIEZO2 ion channels [30, 32, 34]. In the modern mechanotransduction literature, the more relevant gastrointestinal transducer is probably PIEZO2: somatosensory-neuron PIEZO2 signalling has now been shown to regulate gut transit directly [35]. Reaching the 1 µm threshold on the energy-consistent basis requires approximately 158 dB (cf. 135 dB for the pressure-based estimate; Table 3), a level associated with severe acoustic hazards and encountered only in explosive blasts and rocket launches. Even at this extreme, the displacement remains three orders of magnitude below what routine whole-body vibration ($a_{\mathrm{rms}} = 0.5\,\mathrm{m/s^2}$) produces (Table 8).

This framing is, however, *constructive* rather than merely debunking. The model provides the first quantitative explanation for *why* whole-body vibration at 4 Hz to 8 Hz is so effective at producing gastrointestinal effects: it excites a bona fide flexural resonance of the fluid-filled abdominal cavity, producing millimetre-scale tissue displacements that far exceed mechanotransduction thresholds. The epidemiological pattern documented by ISO 2631—peak symptom incidence at 4 Hz to 8 Hz, with a strong dependence on acceleration amplitude—is precisely what a resonance-driven mechanism predicts.

5.3. Alternative Coupling Pathways

The whole-cavity resonance model of §2 treats the abdominal wall as a homogeneous fluid-filled shell. In reality, two additional mechanisms provide alternative pathways for airborne sound to produce tissue displacement: (i) compression of intestinal gas pockets, and (ii) orifice coupling through the upper gastrointestinal tract. We consider each in turn.

5.3.1. Intestinal gas pocket response

The human intestine normally contains 100 mL to 200 mL of gas distributed in pockets with effective radii ranging from 1 cm to 10 cm. Unlike the surrounding tissue, these gas volumes are compressible, and they respond to incident pressure by expanding and contracting their bounding walls. If such local de-

formation is biologically transduced, PIEZO2-dependent gut afferents are the most plausible candidate pathway rather than generic abdominal-wall strain sensing [35].

The classical Minnaert frequency of a spherical gas bubble of radius a in a fluid of density ρ_f is [19]

$$f_M = \frac{1}{2\pi a}\sqrt{\frac{3\gamma P_0}{\rho_f}}, \tag{26}$$

where $\gamma = 1.4$ is the ratio of specific heats and P_0 is the ambient pressure. For physiological pocket radii of 1 cm to 10 cm, this yields Minnaert frequencies in the range 33 Hz to 330 Hz—well above the infrasonic band of interest. At 7 Hz, the frequency ratio $f/f_M \approx 0.02$–0.2, so gas pockets are driven far below resonance and experience no resonant amplification ($H \approx 1.0$).

Nevertheless, in the sub-resonant regime the gas pocket wall displacement is governed by the quasi-static compliance of the trapped gas:

$$\xi_{\text{gas}} \approx \frac{p_{\text{inc}}\, a}{3\gamma P_0}, \tag{27}$$

which scales linearly with pocket radius and is independent of frequency. At 120 dB SPL ($p_{\text{inc}} = 20\,\text{Pa}$) and $a = 2\,\text{cm}$, this yields $\xi_{\text{gas}} \approx 0.94\,\mu\text{m}$—within the PIEZO activation range and roughly seven times the whole-cavity airborne displacement of 0.014 µm at the same SPL. For larger pockets ($a = 5\,\text{cm}$), the displacement rises to approximately 2.4 µm, exceeding the threshold with a modest margin.

Crucially, at 7 Hz the acoustic wavelength (49 m) vastly exceeds the body diameter, so the pressure field inside the body is spatially uniform to within $(ka)^2$ corrections. The gas pockets therefore experience the full incident pressure, without the geometric filtering penalty that suppresses whole-cavity flexural excitation. The mechanism operates as a local pressure-to-displacement transducer embedded within the tissue, rather than as a global shell resonance.

This gas pocket pathway has two important implications. First, it may explain the well-documented inter-individual variability in infrasound sensitivity: bowel gas content varies enormously (50 mL to 500 mL) with diet, gut motil-

ity, and microbiome composition, and individuals with greater gas volume would present more sites for pressure–displacement conversion. Second, the mechanism does not require resonance and operates at any frequency in the sub-wavelength regime, broadening the range of potentially relevant exposures.

5.3.2. Orifice coupling through the upper GI tract

The mouth and nose provide a low-impedance air pathway to the pharynx and—when the esophagus is patent—to the stomach. At 7 Hz, the total path length from mouth to stomach ($L \approx 40\,\mathrm{cm}$) satisfies $kL \approx 0.05 \ll 1$, so pressure is transmitted through the air column with negligible viscothermal attenuation.

With the mouth open and esophagus patent, the pressure reaching the gastric air bubble is approximately equal to the incident pressure, representing a transmission ratio of order unity compared with the $(ka)^2 \approx 1.3 \times 10^{-4}$ penalty for body-surface coupling—an enhancement factor of roughly 7,700×. Compression of the gastric gas bubble then produces wall displacements by the same mechanism described above for intestinal pockets.

However, this pathway is severely limited by anatomy. The esophagus is a collapsible muscular tube that is closed at rest; it opens only during swallowing (approximately 0.5 s to 1.0 s per event), belching, or pathological gastroesophageal reflux. At rest with the mouth open, the air column terminates at the upper esophageal sphincter, and pressure reaches only the supraglottic space. The orifice pathway is therefore *intermittent*, dependent on the state of the upper GI sphincters, and cannot be relied upon as a continuous coupling mechanism.

5.3.3. Assessment of alternative pathways

Table 9 summarises the predicted displacement for each coupling mechanism at 120 dB SPL and 7 Hz.

Even with the most favourable alternative pathway (a large, 5 cm-radius gas pocket at 120 dB), the airborne displacement remains two to three orders of magnitude below what mechanical coupling achieves at the EU Whole-Body

Table 9: Comparison of coupling mechanisms at 120 dB SPL, 7 Hz.

Mechanism	ξ (μm)	Exceeds 1 µm?
Cavity resonance (n=2, airborne)	0.014	No
Gas pocket $a = 2$ cm (airborne)	0.94	Marginal
Gas pocket $a = 5$ cm (airborne)	2.4	Yes
Orifice (open mouth $\rightarrow$ stomach)	$\sim$0.5	Marginal
WBV at 0.5 m/s^2 (mech.)	4,586	Yes
WBV at 1.15 m/s^2 (mech.)	10,549	Yes

Vibration Directive action value of 0.5 m/s^2. The alternative pathways therefore do not alter the principal conclusion of this work—that airborne infrasound is an ineffective excitation route compared with mechanical vibration—but they provide a physically grounded explanation for why rare, sporadic gastrointestinal responses to extreme infrasound (>120 dB) are not entirely implausible, and why such responses exhibit large inter-individual variability.

5.4. Clinical and Occupational Relevance

The coupling disparity has direct implications for occupational exposure standards. ISO 2631-1 specifies frequency-weighted acceleration limits for whole-body vibration, with the highest sensitivity weighting in the 4 Hz to 8 Hz band—precisely the range where our model predicts the $n = 2$ flexural resonance [1], and consistent with von Gierke and Brammer's compilation of organ resonance data [11]. More recent occupational reviews continue to link whole-body vibration exposure with adverse health outcomes across multiple organ systems [41]. The EU Physical Agents (Vibration) Directive 2002/44/EC codifies two thresholds: a daily exposure action value of 0.5 m/s^2 $A(8)$ and an exposure limit value of 1.15 m/s^2 $A(8)$. At the action value, our model predicts a relative abdominal wall displacement of 4.6 mm at resonance (Table 8)—more than 4,000$\times$ the PIEZO activation threshold. The present analysis therefore provides a *mechanistic rationale* for why these particular frequency bands and amplitude

thresholds were empirically selected: they correspond to conditions under which flexural resonance of the abdominal cavity produces tissue strains large enough to activate mechanotransduction pathways.

For airborne infrasound, ISO 7196 defines G-weighting but prescribes no dose–response limits. Our analysis supports this regulatory asymmetry: at environmental and occupational SPL ($\leq 120\,\mathrm{dB}$), airborne coupling to abdominal flexural modes is negligible, and exposure limits based on abdominal resonance effects are unnecessary at these levels.

A potentially valuable extension of this framework is toward personalised vibration exposure assessment. The parametric study (§3) shows that the $n = 2$ eigenfrequency depends sensitively on body habitus: larger abdominal radius (higher BMI) reduces the resonance frequency, while greater muscular tone (higher effective E) increases it. In principle, imaging-derived geometric and viscoelastic parameters [26] combined with the analytical expressions of §2 could yield subject-specific resonance frequency estimates, enabling targeted vibration exposure limits for workers whose body habitus places their resonance frequency in a particularly hazardous band.

5.5. Broader Applications of the Analytical Framework

The brown note provided the motivation for this work, but the analytical framework—a fluid-filled viscoelastic shell, a modal decomposition into breathing and flexural families, and a systematic comparison of airborne versus mechanical coupling efficiency—is considerably more general. Any biological or engineered structure that can be idealised as a compliant enclosure surrounding a dense fluid falls within the scope of the model developed in §2. The two key dimensionless groups that govern the physics, the Helmholtz number ka and the impedance ratio $Z_{\mathrm{wall}}/Z_{\mathrm{fluid}}$, transfer directly; what changes between applications is the frequency range, the material properties, and the geometric aspect ratio of the shell.

Perhaps the most consequential extension is to *primary blast injury.* Explosive detonations generate overpressure pulses with spectral content spanning

10 Hz to 10,000 Hz and peak overpressures of 100 kPa to 1000 kPa—orders of magnitude above the 20 Pa of a 120 dB sound field, but governed by the same impedance-mismatch physics [44, 45, 46]. Gas-containing abdominal organs (bowel, stomach) are clinically the most vulnerable viscera precisely because the gas–tissue interface maximises energy transfer from the blast wave into flexural wall deformation. The coupling comparison of §4 translates directly: replacing the air–tissue impedance ratio with the blast-wave–tissue ratio and adjusting ka for the broadband blast spectrum yields quantitative predictions of wall displacement that could inform threshold models for primary blast bowel injury. Unlike the infrasonic brown-note scenario, blast frequencies place ka near or above unity, so the geometric filtering penalty that rendered airborne coupling negligible at 7 Hz largely disappears—the blast wave *can* efficiently excite flexural modes, which is precisely why blast bowel injury occurs and the brown note does not.

In *fisheries and marine bioacoustics*, the swim bladder of teleost fish is an almost literal instantiation of the present model: a gas-filled viscoelastic shell immersed in a dense fluid [47, 48]. Fisheries sonar operates at frequencies (18 kHz to 200 kHz) chosen to coincide with swim-bladder resonance, where the target strength is maximal. The viscoelastic shell formulation of §2 provides a physics-based alternative to the empirical Love equation for predicting resonance frequency as a function of bladder volume, wall thickness, and depth-dependent ambient pressure. The same framework is directly relevant to environmental impact assessments of naval sonar on marine fauna [49]: organisms possessing gas-filled cavities (swim bladders, lungs of diving mammals) experience tissue strains that the present coupling analysis can bound quantitatively.

The framework also extends to *therapeutic and diagnostic ultrasound*. In high-intensity focused ultrasound (HIFU) and extracorporeal shock-wave lithotripsy, acoustic energy must couple efficiently into a tissue target—the inverse of our brown-note problem, where coupling is deliberately maximised rather than shown to be negligible [50]. The modal decomposition clarifies which spatial harmonics of the incident field deposit energy into flexural versus breathing deforma-

tion of encapsulated structures such as kidney stones, gallstones, or tumour capsules [51]. At therapeutic frequencies (0.5 MHz to 4 MHz), $ka \gg 1$ for organ-scale structures and the long-wavelength penalty vanishes entirely, but for millimetre-scale targets (renal calculi, encapsulated lesions) ka re-enters the order-unity regime where the coupling comparison of §4 again becomes the governing consideration. Separately, in obstetric ultrasonography the amniotic sac constitutes a fluid-filled cavity of dimensions comparable to the adult abdomen; a straightforward reparametrisation of the shell model (thinner wall, lower modulus, smaller radius) would yield resonance predictions relevant to fetal exposure safety assessment [52].

Taken together, these applications illustrate that the coupling-comparison methodology—rather than the brown-note conclusion per se—is the transferable contribution of this work. The conclusion that airborne infrasound cannot produce involuntary defecation is, we concede, of limited industrial utility; the ability to predict *when and why* acoustic–structural coupling to a fluid-filled biological cavity is or is not effective has considerably broader reach.

5.6. Nonlinear Amplitude Effects

The linear model of §2 is valid only for small displacements. To bound the error at occupational vibration amplitudes, we performed a Duffing-type perturbation analysis incorporating membrane stretching and fluid volume conservation, yielding a hardening nonlinearity coefficient $\alpha = 7.07 \times 10^7$.

The onset of appreciable nonlinearity (5 % frequency shift) occurs at a wall displacement of 1091 µm ($A/h = 0.109$). At the EU action value of $0.5\,\mathrm{m/s^2}$, the linear model predicts a peak displacement of 4586 µm; the nonlinear correction reduces this to 2258 µm, a 51 % reduction. The forcing exceeds the jump-phenomenon threshold ($0.5\,\mathrm{m/s^2} > 0.12\,\mathrm{m/s^2}$), so the actual response curve exhibits hysteresis and amplitude-dependent frequency shifts. All modes $n = 2$–6 cross the nonlinearity threshold within the predicted whole-body vibration range.

Crucially, the hardening behaviour means that the linear model *overestimates* peak displacement under mechanical excitation. For the brown note argument, this is a conservative error: the true mechanical coupling is somewhat weaker than the linear prediction, while the airborne displacement ($\sim 0.014\,\mu\text{m}$) remains deep in the linear regime. The 6.6×10^4 coupling disparity therefore represents an upper bound; including the canonical participation correction gives a still-overwhelming ratio of $\sim 3\times10^4$.

5.7. Limitations

We identify the following limitations, grouped by their likely effect on the conclusions.

1. **Linear theory.** The Love–Kirchhoff shell formulation assumes small displacements. The nonlinear analysis of §5.6 shows that occupational vibration amplitudes exceed the 5 % nonlinearity threshold ($A/h > 0.109$), with the hardening response reducing peak displacement by up to 51 % relative to the linear prediction. Because the airborne displacement remains deep in the linear regime, this nonlinearity *reduces* the coupling ratio somewhat but does not alter the qualitative conclusion.
2. **Homogeneous, isotropic wall.** The abdominal wall is a layered anisotropic composite comprising peritoneum, transversalis fascia, internal oblique muscle, external oblique muscle, subcutaneous fat, and skin. Our laminate analysis (not reproduced here) shows that homogenisation introduces less than 5 % error in the effective bending stiffness. The isotropic assumption is a first-order approximation; anisotropy would split the mode degeneracies of the equivalent sphere but would not change the order of magnitude of the eigenfrequencies or the coupling ratio. Recent abdominal-wall mechanics reviews and elastography surveys place the canonical $E = 0.1\,\text{MPa}$ and $\eta = 0.25$ values as representative of contemporary measurement ranges, so the remaining uncertainty is quantitative rather than order-of-magnitude [23, 26].

3. **Free-shell boundary conditions.** The abdomen is partially constrained by the spine posteriorly, the pelvis inferiorly, and the rib cage superiorly. The Rayleigh–Ritz boundary condition study (§3.5) shows that even clamping 25 % of the shell surface shifts f_2 by only 11.4 %, because the fluid added mass (7.4× the shell mass for $n = 2$) dominates the system inertia. This shift is well within the material property uncertainty band, supporting the free-shell approximation. The coupling ratio is unaffected because both excitation pathways address the same modal structure.
4. **Thin-shell approximation.** The wall-thickness-to-radius ratio ($h/R \approx 0.06$) lies near the conventional upper bound of thin-shell theory ($h/R < 0.05$). First-order shear-deformation (Mindlin–Reissner) corrections would reduce the $n = 2$ frequency by approximately 4 % [53], a negligible effect compared with the material-property uncertainty. This limitation is unlikely to affect any conclusions.
5. **Equivalent-sphere geometry.** The oblate spheroidal abdomen is mapped to an equivalent sphere via $R_{\mathrm{eq}} = (a^2c)^{1/3}$. This preserves volume and leading-order curvature but suppresses the equatorial–polar mode splitting that occurs in an oblate spheroid [54]. For a typical aspect ratio $c/a = 0.67$, the splitting is estimated at ∼33 %, which could shift individual mode frequencies but would not change the coupling ratio or the qualitative separation between breathing and flexural mode families.

   More fundamentally, the human abdominal cavity is a *triaxial* ellipsoid (lateral ∼14 cm, craniocaudal ∼16 cm, anteroposterior ∼10 cm), not a spheroid of either kind. The oblate approximation captures the principal feature—anterior–posterior flattening—but any spheroid is a simplification. Crucially, however, the equivalent-sphere flexural frequencies depend on R_{eq} alone: at fixed R_{eq}, varying c/a from 0.3 (extremely oblate) to 3.0 (extremely prolate) changes f_2 by less than 1 %. The geometry enters the coupling ratio only through R_{eq} and the Ritz correction, so the results are robust to the choice of spheroidal idealisation.
6. **Inviscid fluid assumption.** The enclosed fluid is modelled as inviscid.

A viscous correction using the Stokes boundary-layer thickness shows that even at the worst-case mucus viscosity ($\mu = 0.1\,\mathrm{Pa\,s}$), the quality factor decreases by only $\Delta Q = -1.85\%$. The crossover viscosity at which viscous damping would equal structural damping is $281\,\mathrm{Pa\,s}$—the consistency of peanut butter. The inviscid assumption is therefore well justified for any physiologically realistic abdominal fluid.

7. **No internal organ structure.** Visceral organs are homogenised into a single fluid. An effective-medium analysis with solid inclusions (liver, spleen, kidneys) at a volume fraction $\phi = 0.25$ predicts a frequency shift of only $0.4\,\%$ for the $n = 2$ mode. The organ tissue bulk modulus ($K_{\mathrm{tissue}} \approx K_{\mathrm{water}}$) produces zero effective bulk modulus change, and at sub-percolation volume fractions the isolated inclusions cannot transmit shear. The homogeneous fluid model is therefore adequate for the present purposes.

8. **Gas pocket model simplifications.** The gas pocket analysis (§5.3) treats intestinal gas as spherical bubbles in an infinite homogeneous fluid, neglecting the cylindrical geometry of the gut lumen, viscoelastic wall loading, and the surrounding organ structure. These simplifications likely *overestimate* gas pocket displacement, because intestinal wall mass loading and stiffness would reduce the effective compliance. A coupled gas–tissue finite-element model is needed to refine these estimates.

9. **Mechanotransduction thresholds.** The $0.5\,\mu\mathrm{m}$ to $2.0\,\mu\mathrm{m}$ PIEZO activation range [30, 32, 34] derives from cell-culture and membrane-tension experiments and may not directly translate to tissue-scale effects, where strain concentrations around cells are modulated by the extracellular matrix. In gastrointestinal tissue the relevant transduction pathway is also likely to involve PIEZO2-expressing afferents rather than PIEZO1 alone [35]. This limitation introduces uncertainty in the absolute threshold but does not affect the *relative* comparison between airborne and mechanical coupling.

In summary, the identified limitations affect the absolute values of predicted frequencies and displacements—in several cases substantially—but none alter the central finding that airborne acoustic coupling to abdominal flexural modes is 10^4–10^5 times weaker than mechanical coupling under occupationally relevant conditions.

5.8. Proposed Experimental Validation

We propose a two-stage experimental programme to validate the key predictions of the analytical model. The first stage employs a tissue-mimicking phantom: an oblate spheroidal silicone shell ($a = 0.18\,\mathrm{m}$, $c = 0.12\,\mathrm{m}$, $h = 10\,\mathrm{mm}$) filled with degassed water, matching the canonical geometry of §2. The shell material is platinum-cure silicone (Ecoflex 00-30 to Dragon Skin 10A), whose Young's modulus spans 0.07 MPa to 0.15 MPa—bracketing the relaxed abdominal wall estimate of $E = 0.1\,\mathrm{MPa}$. By casting phantoms at several mixing ratios, the $f_2 \propto \sqrt{E}$ dependence can be tested directly. Modal frequencies are measurable with an instrumented impact hammer or a low-frequency shaker sweep, and mode shapes can be reconstructed with a scanning laser vibrometer (Polytec PSV-500). The predicted $n = 2$ frequency for the recommended Ecoflex 00-30 formulation is 2.5 Hz; for Dragon Skin 10A, 3.6 Hz; the model predicts 4.0 Hz for the canonical human case.

The critical measurement is the coupling-ratio comparison. By mounting the phantom on an electrodynamic shaker (mechanical pathway) and separately exposing it to a subwoofer-generated infrasound field at 120 dB (airborne pathway), the displacement ratio $\xi_{\mathrm{mech}}/\xi_{\mathrm{air}}$ can be measured in a single experimental session. The model predicts ratios of order 10^4–10^5 across the silicone formulation range (Table 9). Non-detection of the airborne response at the vibrometer noise floor ($\sim 0.01\,\mu\mathrm{m}$) would itself constitute a positive validation of the weak-coupling prediction. The total cost of consumables (silicone, molds, sensors) is estimated at less than US$ 3 100, assuming access to shared laboratory equipment (shaker, laser vibrometer, DAQ).

The second stage extends to in vivo human subjects. Subject-specific ab-

dominal wall moduli would be obtained via magnetic resonance elastography or shear-wave ultrasound elastography, and whole-body vibration transmissibility measured on an ISO 2631-compliant platform at sub-action-value amplitudes ($0.3\,\mathrm{m/s^2}$). A sample of 15 subjects, spanning a range of body habitus, is sufficient (power $= 0.80$, $\alpha = 0.05$) to detect a paired-test effect size $d = 1.0$ between measured and predicted peak-transmissibility frequencies. This programme requires Institutional Review Board approval and is estimated at 8–12 months, with the phantom study serving as a prerequisite validation of the model physics under controlled conditions.

6. Conclusions

The central result of this study is an upper-bound coupling disparity of approximately 6.6×10^4 between mechanical whole-body vibration and airborne acoustic excitation of the human abdominal cavity for 120 dB SPL airborne loading and $a_{\mathrm{rms}} = 0.1\,\mathrm{m/s^2}$ RMS whole-body vibration under the SDOF idealisation ($\Gamma_2 = 1$). This single quantity, derived from first-principles shell theory and energy-consistent reciprocity analysis, resolves a long-standing empirical asymmetry: occupational whole-body vibration at 4 Hz to 8 Hz causes well-documented gastrointestinal effects, while airborne infrasound at comparable frequencies and extreme intensities does not.

The disparity has a clean physical origin. The abdominal cavity possesses two distinct mode families: a breathing mode ($n = 0$) near 2500 Hz, governed by the fluid bulk modulus and therefore irrelevant at infrasonic frequencies; and flexural modes ($n \geq 2$) in the 4 Hz to 10 Hz range that match ISO 2631 abdominal transmissibility data. Airborne sound at these frequencies has a wavelength exceeding 85 m, placing the abdomen deep in the Rayleigh regime ($ka \approx 0.01$). The $(ka)^n$ geometric penalty, compounded by the air–tissue impedance mismatch, reduces the effective driving pressure by four orders of magnitude. An energy-consistent analysis confirms an absorption efficiency of $\sim 10^{-14}$: the shell is, for practical purposes, acoustically transparent. Reaching the mechan-

otransduction threshold via airborne infrasound alone would require approximately 158 dB on the energy-consistent basis (135 dB using the less physical pressure-based estimate)—a level at which structural damage to the middle ear and other severe acoustic hazards would present more immediate concerns than any intestinal indignity.

Mechanical vibration, transmitted through the skeleton and abdominal wall, couples directly and, for the headline comparison (120 dB SPL airborne versus $a_{\mathrm{rms}} = 0.1\,\mathrm{m/s^2}$ RMS WBV with $\Gamma_2 = 1$), produces displacements approximately 6.6×10^4 times larger; including the participation correction leaves $\sim 3 \times 10^4$. This disparity is not a subtle biomechanical effect but a straightforward consequence of long-wavelength acoustics and impedance physics. Monte Carlo uncertainty quantification using the pressure-based airborne upper bound yields a ratio spanning roughly one and a half orders of magnitude, with a 90% credible interval of $[2 \times 10^3,\ 3 \times 10^4]$; elastic modulus accounts for 86% of total variance.

The brown note, then, is a half-truth. The resonance it invokes is genuine; the mechanism it implies is not. What popular culture attributes to airborne sound is, in reality, the province of mechanical vibration—a distinction that matters considerably for occupational health standards and rather less for horror film screenwriters. Whether localised compressible inclusions (intestinal gas pockets) provide a secondary transduction pathway at extreme intensities remains an open question; they operate by quasi-static compliance, not by cavity resonance, and are explored separately.

Future work should validate the boundary condition assumptions through finite element analysis of anatomically realistic geometries, investigate the role of bowel gas pockets as localised acoustic transducers, and extend the model into the nonlinear regime relevant to high-amplitude mechanical excitation. In the meantime, concertgoers may rest assured—or, more precisely, remain seated—without fear of acoustic indignity.

References

[1] International Organization for Standardization, ISO 2631-1:1997 — mechanical vibration and shock — evaluation of human exposure to whole-body vibration, Standard (1997).

[2] M. J. Griffin, Handbook of Human Vibration, Academic Press, 1990.

[3] N. J. Mansfield, Human Response to Vibration, CRC Press, 2005.

[4] G. C. Mohr, J. N. Cole, E. Guild, H. E. von Gierke, Effects of low frequency and infrasonic noise on man, Aerospace Medicine 36 (9) (1965) 817–824.

[5] A. Savage, J. Hyneman, Brown note, *MythBusters*, Season 3, Episode 1 (overall Episode 25), first aired 16 February 2005 on Discovery Channel, brown-note myth tested and busted (2005).
URL `https://mythresults.com/episode25`

[6] G. Leventhall, What is infrasound?, Progress in Biophysics and Molecular Biology 93 (1–3) (2007) 130–137. `doi:10.1016/j.pbiomolbio.2006.07.006`.

[7] S. Raufer, S. F. Masud, H. H. Nakajima, Infrasound transmission in the human ear: Implications for acoustic and vestibular responses, Journal of the Acoustical Society of America 144 (1) (2018) 332–342. `doi:10.1121/1.5046523`.

[8] T. Parker, M. Stone, World wide recorder concert ("the brown noise"), *South Park*, Season 3, Episode 17, first aired 12 January 2000 on Comedy Central, directed by Eric Stough; production code 317 (2000).
URL `https://en.wikipedia.org/wiki/The_Brown_Noise`

[9] V. Gavreau, R. Condat, H. Saul, Infra-sons: générateurs, détecteurs, propriétés physiques, effets biologiques, Acustica 17 (1) (1966) 1–10.

[10] H. E. von Gierke, Effects of infrasound on man, Technical Note TN D-7620, NASA (1974).

[11] H. E. von Gierke, A. J. Brammer, Effects of shock and vibration on humans, in: C. M. Harris, A. G. Piersol (Eds.), Harris' Shock and Vibration Handbook, 5th Edition, McGraw-Hill, 2002, Ch. 42.

[12] V. Tandy, T. R. Lawrence, The ghost in the machine, Journal of the Society for Psychical Research 62 (851) (1998) 360–364.

[13] G. Leventhall, A review of published research on low frequency noise and its effects, Tech. rep., Department for Environment, Food & Rural Affairs (DEFRA), London, UK (2003).

[14] J. Altmann, Acoustic weapons — a prospective assessment, Occasional Paper 22, Center for Security Studies and Conflict Research, ETH Zurich (1999).

[15] J. R. Jauchem, M. C. Cook, High-intensity acoustics for military nonlethal applications: A lack of useful systems, Military Medicine 172 (2) (2007) 182–189. doi:10.7205/MILMED.172.2.182.

[16] I. van Kamp, F. van den Berg, Health effects related to wind turbine sound: An update, International Journal of Environmental Research and Public Health 18 (17) (2021) 9133. doi:10.3390/ijerph18179133.

[17] L. Ascone, et al., A longitudinal, randomized experimental pilot study to investigate the effects of airborne infrasound on human mental health, cognition, and brain structure, Scientific Reports 11 (2021) 3190. doi:10.1038/s41598-021-82203-6.

[18] M. C. Junger, D. Feit, Sound, Structures, and Their Interaction, 2nd Edition, MIT Press, 1986.

[19] H. Lamb, On the Vibrations of an Elastic Sphere, Vol. s1-13, Proceedings of the London Mathematical Society, 1882.

[20] K. A. Kuo, H. E. M. Hunt, J. R. Lister, Small oscillations of a pressurized, elastic, spherical shell: Model and experiments, Journal of Sound and Vibration 359 (2015) 168–178. doi:10.1016/j.jsv.2015.08.021.

[21] A. Ghaheri, M. T. Ahmadian, F. Fallah, Free vibration analysis of a fluid-filled functionally graded spherical shell subjected to internal pressure, Acta Mechanica 233 (2022) 3095–3112. doi:10.1007/s00707-022-03262-y.

[22] F. Alijani, M. Amabili, Non-linear vibrations of shells: A literature review from 2003 to 2013, International Journal of Non-Linear Mechanics 58 (2014) 233–257. doi:10.1016/j.ijnonlinmec.2013.09.012.

[23] C. R. Deeken, S. P. Lake, Mechanical properties of the abdominal wall and biomaterials utilized for hernia repair, Journal of the Mechanical Behavior of Biomedical Materials 74 (2017) 411–427. doi:10.1016/j.jmbbm.2017.05.008.

[24] S. Spadoni, et al., Numerical modeling of the abdominal wall biomechanics and experimental analysis for model validation, Frontiers in Bioengineering and Biotechnology 12 (2024) 1472509. doi:10.3389/fbioe.2024.1472509.

[25] J. Novák, et al., Intra-abdominal pressure correlates with abdominal wall tension during clinical evaluation tests, Clinical Biomechanics 88 (2021) 105426. doi:10.1016/j.clinbiomech.2021.105426.

[26] I. Sack, Magnetic resonance elastography from fundamental soft-tissue mechanics to diagnostic imaging, Nature Reviews Physics 5 (2023) 223–242. doi:10.1038/s42254-022-00543-2.

[27] K. J. Parker, M. M. Doyley, D. J. Rubens, Imaging the elastic properties of tissue: The 20 year perspective, Physics in Medicine and Biology 56 (1) (2011) R1–R29.

[28] M. Heil, Large post-buckling deformations of cylindrical shells conveying viscous flow, Journal of Fluids and Structures 10 (6) (1996) 565–599. doi:10.1006/jfls.1996.0039.

[29] S. Kitazaki, M. J. Griffin, Resonance behaviour of the seated human body and effects of posture, Journal of Biomechanics 31 (2) (1998) 143–149.

[30] B. Coste, J. Mathur, M. Schmidt, T. J. Earley, S. Ranade, M. J. Petrus, A. E. Dubin, A. Patapoutian, Piezo1 and piezo2 are essential components of distinct mechanically activated cation channels, Science 330 (6000) (2010) 55–60.

[31] O. A. Godin, Sound scattering and radiation suppression by pressurized spherical shells, Journal of the Acoustical Society of America 154 (5) (2023) 3223–3236. doi:10.1121/10.0022416.

[32] A. H. Lewis, J. Grandl, Mechanical sensitivity of piezo1 ion channels can be tuned by cellular membrane tension, eLife 4 (2015) e12088. doi:10.7554/eLife.12088.

[33] K. Saotome, et al., Structure of the mechanically activated ion channel piezo1, Nature 554 (2018) 481–486. doi:10.1038/nature25453.

[34] B. Xiao, Mechanisms of mechanotransduction and physiological roles of PIEZO channels, Nature Reviews Molecular Cell Biology 25 (11) (2024) 886–903. doi:10.1038/s41580-024-00773-5.

[35] M. R. Servin-Vences, et al., PIEZO2 in somatosensory neurons controls gastrointestinal transit, Cell 186 (16) (2023) 3562–3577. doi:10.1016/j.cell.2023.07.006.

[36] S. Gupta, et al., Effect of the acoustic impedance mismatch at the bone-soft tissue interface as a function of frequency, IEEE Transactions on Ultrasonics, Ferroelectrics, and Frequency Control 68 (5) (2021) 1653–1663. doi:10.1109/TUFFC.2020.3043893.

[37] G. Zheng, Y. Qiu, M. J. Griffin, Vertical and dual-axis vibration of the seated human body: Nonlinearity, cross-axis coupling, and associations between resonances, Journal of Sound and Vibration 331 (26) (2012) 5880–5894. doi:10.1016/j.jsv.2012.07.029.

[38] K. Gao, et al., Finite element modeling and parameter identification of the seated human body exposed to vertical vibration, Biomechanics

and Modeling in Mechanobiology 20 (2021) 1789–1803. doi:10.1007/s10237-021-01481-1.

[39] H. Seidel, R. Heide, Long-term effects of whole-body vibration: A critical survey of the literature, International Archives of Occupational and Environmental Health 58 (1) (1986) 1–26. doi:10.1007/BF00378536.

[40] T. Ishitake, Y. Miyazaki, R. Noguchi, H. Ando, T. Matoba, Evaluation of frequency weighting (ISO 2631-1) for acute effects of whole-body vibration on gastric motility, Journal of Sound and Vibration 253 (1) (2002) 31–36. doi:10.1006/jsvi.2001.4263.

[41] K. Krajnak, Health effects associated with occupational exposure to hand-arm or whole body vibration, Journal of Toxicology and Environmental Health, Part B 21 (5) (2018) 320–334. doi:10.1080/10937404.2018.1557576.

[42] H. Møller, C. S. Pedersen, Hearing at low and infrasonic frequencies, Noise & Health 6 (23) (2004) 37–57.

[43] R. Dong, et al., Study on the biodynamic characteristics and internal vibration behaviors of a seated human body, Biomechanics and Modeling in Mechanobiology 23 (2024) 1449–1468. doi:10.1007/s10237-024-01849-z.

[44] J. H. Stuhmiller, Y. Y. Phillips, D. R. Richmond, The Physics and Mechanisms of Primary Blast Injury, Textbooks of Military Medicine, Office of the Surgeon General, Borden Institute, 2008.

[45] C. Owers, J. L. Morgan, J. P. Garner, Abdominal trauma in primary blast injury, British Journal of Surgery 98 (2) (2011) 168–179. doi:10.1002/bjs.7268.

[46] M. A. Mayorga, The pathology of primary blast overpressure injury, Toxicology 121 (1) (1997) 17–28. doi:10.1016/S0300-483X(97)03652-4.

[47] C. Feuillade, R. W. Nero, A viscous–elastic swimbladder model for describing enhanced-frequency resonance scattering from fish, Journal of the Acoustical Society of America 103 (6) (1998) 3245–3255. doi:10.1121/1.423077.

[48] R. H. Love, Resonant acoustic scattering by swimbladder-bearing fish, Journal of the Acoustical Society of America 64 (2) (1978) 571–580. doi:10.1121/1.382009.

[49] B. L. Southall, A. E. Bowles, W. T. Ellison, J. J. Finneran, R. L. Gentry, C. R. Greene, Jr., D. Kastak, D. R. Ketten, J. H. Miller, P. E. Nachtigall, W. J. Richardson, J. A. Thomas, P. L. Tyack, Marine mammal noise exposure criteria: Initial scientific recommendations, Aquatic Mammals 33 (4) (2007) 411–521. doi:10.1578/AM.33.4.2007.411.

[50] G. ter Haar, C. Coussios, High intensity focused ultrasound: Physical principles and devices, International Journal of Hyperthermia 23 (2) (2007) 89–104. doi:10.1080/02656730601186138.

[51] M. R. Bailey, V. A. Khokhlova, O. A. Sapozhnikov, S. G. Kargl, L. A. Crum, Physical mechanisms of the therapeutic effect of ultrasound (a review), Acoustical Physics 49 (4) (2003) 369–388. doi:10.1134/1.1591291.

[52] F. A. Duck, Medical and Non-Medical Protection Standards for Ultrasound and Infrasound, Elsevier, 1999, review of exposure standards for diagnostic ultrasound.

[53] W. Soedel, Vibrations of Shells and Plates, 3rd Edition, Marcel Dekker, 2004. doi:10.1201/9780203026304.

[54] A. W. Leissa, Vibration of Shells, NASA, 1973, nASA SP-288.

Data availability

All analytical models and computational scripts used in this study are available in the accompanying repository.

CRediT authorship contribution statement

Jonathan Mace: Conceptualization, Methodology, Supervision, Writing – Review & Editing. **Brian R. Mace**: Writing – Review & Editing, Validation. **GitHub Copilot CLI**: Software, Formal Analysis, Investigation, Data Curation, Writing – Original Draft, Visualization.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.